

Ali Gill

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Toronto, Ontario, Canada

SUMMARY

BSc (Hons) candidate at the University of Toronto (History and Philosophy of Science and Technology | Mathematics, 2027). Offensive security background with CPTS certification, alongside AI/ML engineering experience in RAG, LLM applications, and agentic systems. Particularly interested in **AI security**: adversarial machine learning, prompt injection, and securing LLM-based systems.

EDUCATION

University College - University of Toronto,

Sept 2023 - Sept 2027

BSc (Hons). History and Philosophy of Science and Technology | Mathematics (Double Major)

Computer Science (Minor)

Relevant Coursework: Software Design, Programming on the Web, Software Tools and Systems Programming, Statistical Reasoning and Data Science, Mathematical Expression and Reasoning for Computer Science, Calculus, Linear Algebra

Aitchison College, GCE O & A levels

Lahore, Pakistan | Sept 2010 - June 2022

RELEVANT EXPERIENCE

Enterprise Resource Planning Technical Consultant - Systems Limited, Internship

Lahore, Pakistan | Aug 2026 - Sep 2026

- Worked with the technical consulting team on Microsoft Dynamics ERP, supporting the technical workstream of an enterprise implementation.
- Gained hands-on exposure to ERP configuration and customization, and to how business processes map onto system modules.
- Worked with the AI capabilities layered over the ERP platform, including Copilot-style assistance and AI-driven insights over operational data.
- Examined where machine learning fits into ERP workflows, from forecasting and anomaly detection on transactional data to automating routine data entry.

AI/ML Engineering Intern - Allied Bank Limited, Internship

Lahore, Pakistan | Jun 2026 - Aug 2026

- Worked with the Analytics & AI team to build AI/ML solutions for business use cases, focusing on Retrieval-Augmented Generation (RAG), LLM applications, and Agentic AI.
- Designed intelligent workflows integrating internal business data with context-aware AI agents to support automation and decision-making across banking operations.
- Built retrieval pipelines that grounded model output in the bank's own document sources rather than general model knowledge.
- Used vector embeddings and semantic search to pull relevant context out of unstructured internal documents at query time.
- Evaluated agent behaviour on multi-step tasks, tuning prompts and tool definitions to make outputs more reliable.

Database Design Intern - Code District, Internship

- Worked with the database team on a client project, participating in database design workshops and gaining hands-on exposure to data modelling and database development.
- Built a foundational understanding of database architecture and design practices.

Software Development Intern - Agilitize, Internship

- Gained practical exposure to the software development lifecycle through requirement gathering and test case development.
- Analyzed requirements, communicated with stakeholders, and developed structured test cases for the product under test.

PROJECTS

Security Writeups & Notes - Technical Documentation - Medium: [@0xikon](#), Notes: [0xikon.gitbook.io](#)

- Publish HackTheBox writeups and a CPTS certification review on Medium, covering enumeration to privilege escalation.
- Maintain a companion GitBook knowledge base (60+ pages) on web, wireless, Active Directory, and Linux exploitation.

CMS - Software Design, University of Toronto - Team of 4

- Built a multi-user desktop chat application in Java with authentication, chatroom creation/discovery, and messaging.
- Structured the codebase with Clean Architecture and a DAO persistence layer, applying SOLID principles throughout.

LoyaLoop - Loyalty Rewards Management System - React, NodeJS/Express, Prisma, JWT, University of Toronto

- Built a full-stack loyalty-rewards platform (React 19 + Material-UI, Node.js/Express REST API, Prisma over SQLite).
- Implemented JWT role-based access control across four roles, a promotion engine, and QR-code point redemption.
- Added a real-time WebSocket analytics dashboard; hardened against XSS, CSRF, and SQLi; deployed on Railway.

CERTIFICATIONS/TRAINING

Certifications: CompTIA Linux+, Security+, Certified Penetration Testing Specialist (CPTS), Junior Penetration Tester (eJPT), Practical OSINT Research Professional (PORP) [HTB, THM profiles]

Training: Stanford/DeepLearning.AI Machine Learning Specialization, Google Cybersecurity Specialization, MIT Cybersecurity for Critical Urban Infrastructure, Harvard CS50x, Stanford Advanced Cybersecurity Program

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash, JavaScript (HTML/CSS)

Security & Pentesting: Burp Suite, Nmap, Metasploit, Wireshark, BloodHound, Ligolo-ng, Hashcat, ffuf/gobuster, sqlmap, Active Directory exploitation, report writing

ML & Data: NumPy, pandas, scikit-learn, Jupyter, TensorFlow

Infrastructure: Linux (Arch/Debian), Docker, Git/GitHub, REST APIs, virtualization